In [ ]:

In [109…
```
import cv2
import requests
import numpy as np
import matplotlib.pyplot as plt
def get_img(url):
    resp = requests.get(url)
    return cv2.imdecode(np.frombuffer(resp.content, np.uint8), cv2.IMREAD_GRAYSCALE

img = get_img("https://raw.githubusercontent.com/ashish-kmr/BTP_inpaint/master/came

img
```

Out[109… ndarray (512, 512) show data



In [110…
```
cv2.imwrite('/content/output.jpg', img)
```

Out[110… True

Tasks

1. Increase the size of the image.
2. Rotate the image by 120 degrees.
3. Perform Sheer operations over the image.
4. Create a Negative Image
5. Log with (choose an appropriate constant for your image)
6. Power law (choose a value for gamma that increases the contrast)

In [111…
```
# 1.    Increase the size of the image.
cv2.resize(img, (784, 784))
```

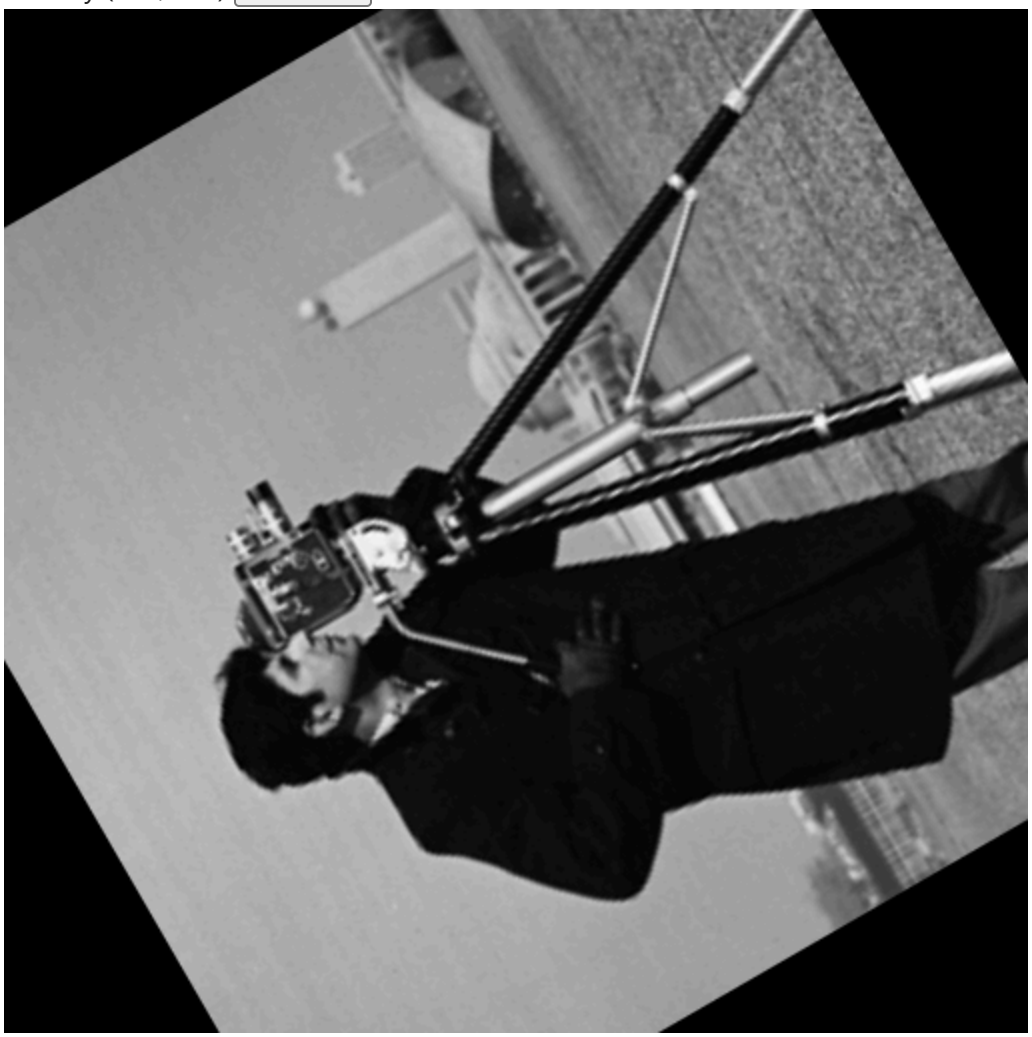
Out[111…
ndarray (784, 784) show data



In [112…
```
# 2.    Rotate the image by 120 degrees.
h, w = img.shape[:2]
```

```
M = cv2.getRotationMatrix2D((w / 2, h / 2), 120, 1)
rotated = cv2.warpAffine(img, M, (w, h))
rotated
```

Out[112… ndarray (512, 512) show data



In [113…

```
# 3.    Perform Sheer operations over the image.
h, w = img.shape[:2]
M = np.float32([[1, 0.5, 0], [0, 1, 0]])
sheared = cv2.warpAffine(img, M, (w, h))
sheared
```

Out[113… ndarray (512, 512) show data



In [114…

```
# 1.    Create a Negative Image
negative = cv2.bitwise_not(img)
negative
```

Out[114… ndarray (512, 512) show data



In [121…

```
img_float = img.astype(np.float32)

c = 30
log_img = c * np.log(1 + img_float)

log_img = log_img.astype(np.uint8)
log_img
```

Out[121… ndarray (512, 512) show data



In [116…
```
fig, axs = plt.subplots(2,5,figsize=(15,6)); [((axs.flat[i].imshow(np.clip(c*np.log
```



In [122…
```
gamma = 0.5
power_img = np.uint8(255 * (img / 255.0) ** gamma)
power_img
```

Out[122… ndarray (512, 512) show data



In [118…

```
fig, axs = plt.subplots(2,5,figsize=(15,6)); [((axs.flat[i].imshow(255*(img/255.0)*
```

